Exp No: 30 – RTOS Task Scheduling

AIM:

To simulate Round-Robin task scheduling in SystemC and observe the periodic execution of multiple tasks.

PROCEDURE:

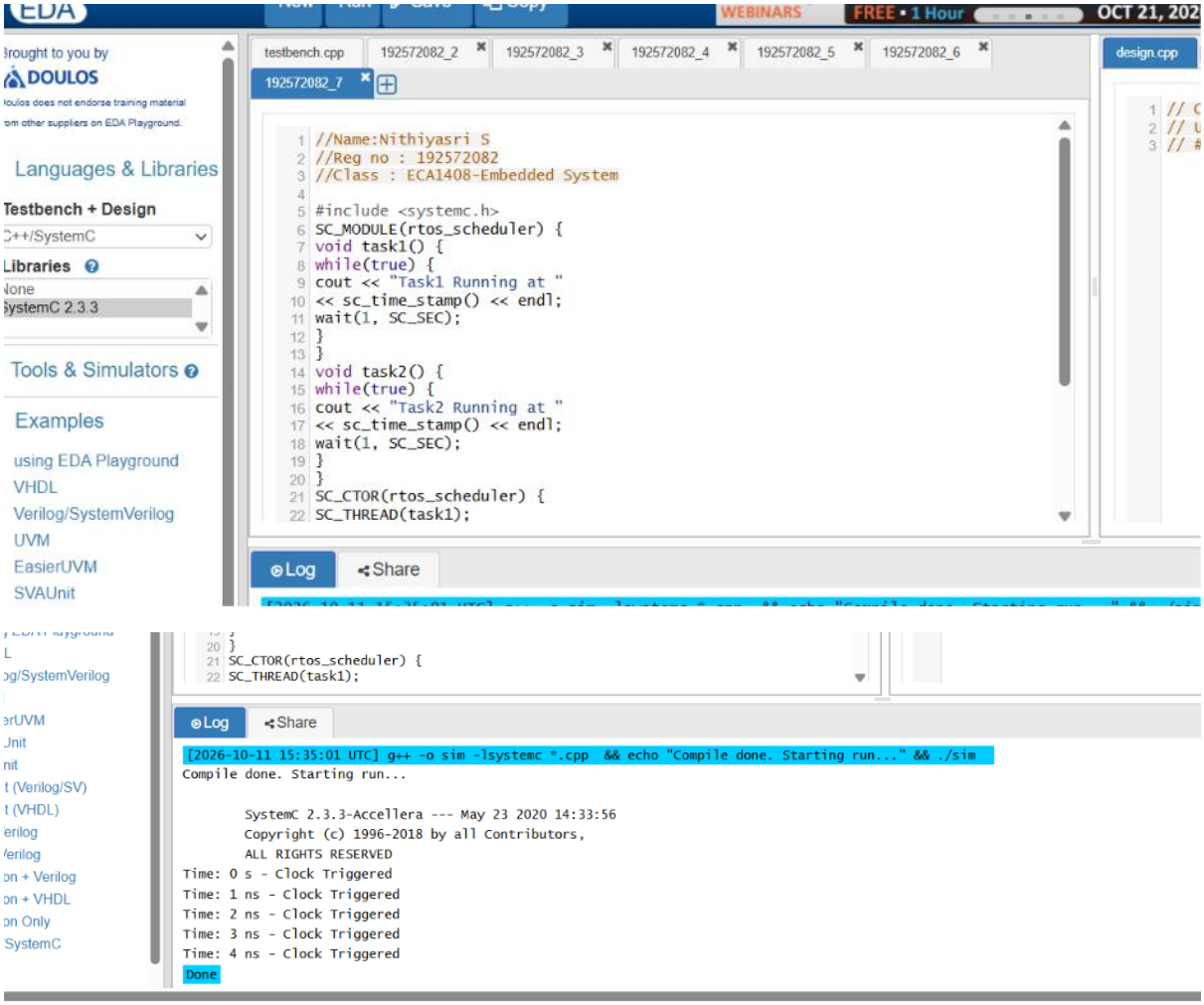
1. Create an RTOS scheduler module using SC_MODULE.
2. Define two concurrent tasks, task1() and task2(), using separate functions.
3. Register both tasks using SC_THREAD in the constructor.
4. Use wait(1, SC_SEC) in each task to simulate periodic execution at one-second intervals.
5. Run the simulation for 6 seconds and observe the execution timestamps of both tasks.

PROGRAM:

```
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() {
while(true) {
cout << "Task1 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
void task2() {
while(true) {
cout << "Task2 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
```

```
SC_THREAD(task2);
}
};

int sc_main(int argc, char* argv[]) {
rtos_scheduler obj("Scheduler");
sc_start(6, SC_SEC);
return 0;
}
```



RESULT:

Thus, RTOS task scheduling was simulated using SystemC, and the periodic execution of two concurrent tasks was verified.